

Autumn 2025

Security and Privacy

Threat analysis of hacktivist attacks on EU infrastructure

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August 31, 2025

1 Introduction

This report analyses activities on Telegram groups used by hacktivists targeting critical infrastructure in EU and Nordic countries. The main research questions are:

- Which EU + Nordic countries were targeted?
- Which critical infrastructure sectors were attacked?
- Which security properties (confidentiality, integrity, availability) were affected?

2 Dataset Description

- Source: Telegram group dataset (messages provided in the assignment) as cvs file.
- Size: 3004 messages.
- Structure: Each record contains id, timestamp and message.

3 Methodology

3.1 Text Preprocessing

- Lowercasing, removal of special characters.

```
# Lowercase all Text
df['Text'] = df['Text'].str.lower()
# Remove \n and \r
df['Text'] = df['Text'].str.replace(r'[\n\r]', ' ', regex=True)
```

- Handling spelling errors and variations.
Spelling errors of countries is partially solved by catching countries by emoji or base of the country name, such as 'finn' for Finland.

3.2 Country Identification

- Created mapping of country names and flags to standard country codes.
I noticed, that mostly, there is mentioned country of attack as very first.
Country could be mentioned as emoji, therefor I created regex to map emoji flag to country name:

```
FLAG_TEXT_TO_COUNTRY = {
    "<emoji of DK flag>": "Denmark",
    "<emoji of FIN flag>": "Finland",
    ...
}
```

If flag was not found and matched, try text variations:

```
...
    "danmark": "Denmark",
    "danish": "Denmark",
    "finn": "Finland",
    ...
```

- Applied regex search on message texts.
Creating new column in pandas dataframe, with return value (country that matched as fist (by emoji or flag in this order)) of function that will apply regex on message.

```
df['First_flag_country'] = df['Text'].apply(detect_country_from_text)
```

3.3 Sector Identification

- Used keyword lists based on EU list of essential services but slightly adjusted to catch messages in different formats.

```
"Energy sector": re.compile(
    r"(power plant|power station|substation|electricity|wind farm|
    solar|photovoltaic|hydroelectric|geothermal|battery|
    transformer|grid operator|tso|dso)"
),
"Transport sector": re.compile(
    r"(airport|runway|railway|railroad|train|metro|subway|tram|
    seaport|harbor|harbour|logistics|freight|highway|toll)"
),
...
```

- Applied regex search on message texts.
Here, new column 'sector' was created, where return of function that is applying regex above is saved as value.

```
df['sector'] = df['Text'].apply(detect_sectors)
```

3.4 Security Property Identification

CIA triad:

- **Confidentiality** - message referring to data leaks
- **Integrity** - message referring to taking over system configurations and internal access
- **Availability** - message referring to taking down website or system - so nobody can have access

These categories allowed us to tag each relevant message in the dataset according to which property of the CIA triad was targeted.

4 Results

4.1 Targeted Countries

- List of countries with frequency counts.
- Visualization (see Figure 1)

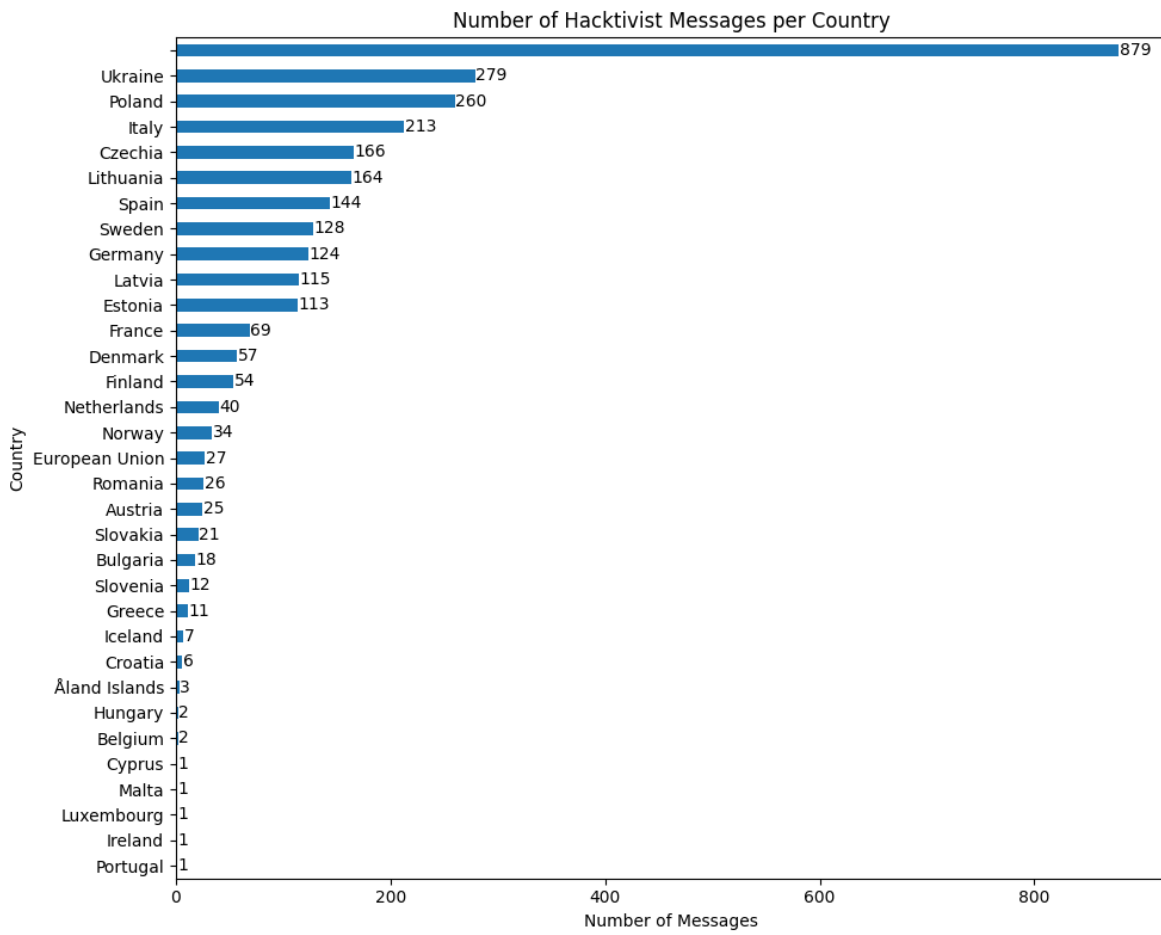


Figure 1: Bar chart with counts of unique messages matched with country including not matched count

4.2 Attacked Sectors

- Sectors mentioned
- Visualization (see Figure 2)

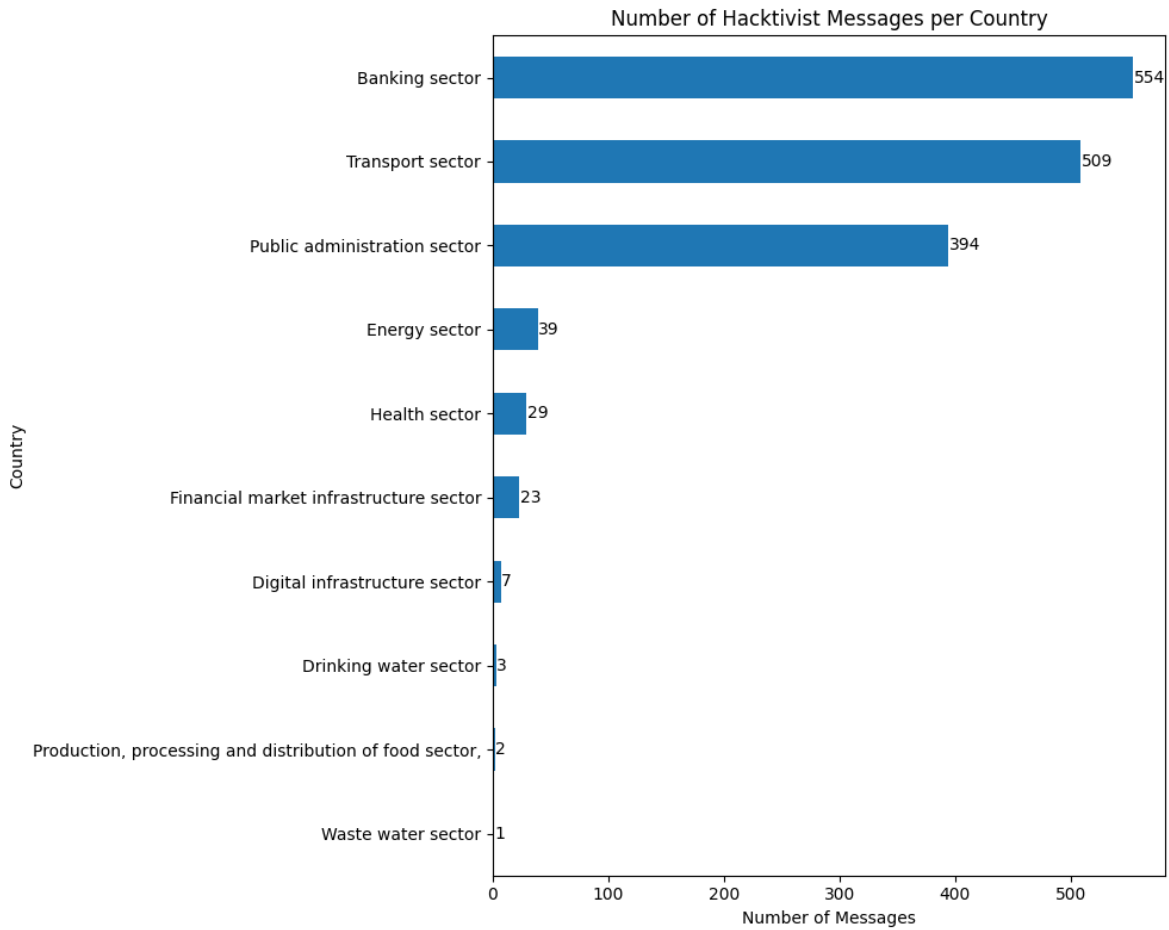


Figure 2: Bar chart with counts of unique messages matched with sector

4.3 Security Properties

CIA triad:

- **Confidentiality** Attackers are informing that they leaked sensitive data
 - Example id=2692: “...databases were leaked, system and database backups were deleted,...”
- **Integrity** Attackers are informing that they hacked into Spanish system
 - Example id=2692: “...hacked into Cde Cantos (Spain)...”
- **Availability** Here attacker is informing that they took down website
 - Example id=2071: “...Our response will be more decisive: we have blocked access to important subdomains of the SITA website...”

5 Discussion and Limitations

- Not matching all messages
My approach might miss countries, that were misspelled or mentioned in different format or

language. Investigating and debugging could bring more identifications.

- Some sectors hard to classify.

My approach is not identifying all possible sector variations - for example banking transport sector attack could be written by words that I am currently not catching in regex. Investigating and debugging could bring more identifications.

- Falsely matched country or sectors

Since I am matching first match, it could happen that first match is attacker country or some country mentioned but not being attacked therefor I might identify some messages wrongly.

6 Conclusion

- Summary of findings.

Based on my results I can say that most of the attacks were on Ukraine, then Poland.

This might be because of activity of Russian hackers that are mainly attacking Ukraine and then countries that are helping Ukraine the most.

Regarding sectors, banking sector is mostly attacked, followed by transport sector.

- Future work: Work on better regex or train NLP model to classify better.